A Context-Aware Assistance Framework for Implicit Interaction with an Augmented Human

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Abstract. The automotive industry is currently facing massive challenges. Shorter product life cycles together with mass customization lead to a high complexity for manual assembly tasks. This induces the need for effective manual assembly assistances which guide the worker faultlessly through different assembly steps while simultaneously decrease their completion time and cognitive load. While in the literature a simulation-based assistance visualizing an augmented digital human was proposed, it lacks the ability to incorporate knowledge about the context of an assembly scenario through arbitrary sensor data. Within this paper, a general framework for the modular acquisition, interpretation and management of context is presented. Furthermore, a novel context-aware assistance application in augmented reality is introduced which enhances the previously proposed simulation-based assistance method by several context-aware features. Finally, a preliminary study (N = 6) is conducted to give a first insight into the effectiveness of context-awareness for the simulation-based assistance with respect to subjective perception criteria. The results suggest that the user experience is improved by context-awareness in general and the developed context-aware features were overall perceived as useful in terms of error, time and cognitive load reduction as well as motivational increase. However, the developed software architecture offers potential for improvement and future research considering performance parameters is mandatory.

Keywords: Augmented reality · Context-awareness · Human computer interaction · Assistance · Human simulation

1 Introduction

Context-aware assistance applications can be a promising possibility to cope with complexity or novelty of manual tasks. Due to the incorporation of environmental and personal knowledge, individualized instructions are tailored to

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the specific needs and goals of the user and environmental requirements. Therefore, such assistance systems for manual tasks are widely investigated, independent of the specific domain [9,13,16]. With regard to the massive challenges the automotive industry is faced with, context-aware worker assistances gain importance. Shorter product life cycles together with mass customization lead to a high complexity for manual assembly tasks [11]. This induces the need for effective and ergonomic manual assembly assistances and manual assembly trainings, which guide the worker faultlessly through different assembly steps while simultaneously, decrease their completion time and cognitive load. In particular, the relevance of augmented reality (AR) assistance methods increase, due to the possibility of real time interaction, the combination of the real environment with virtual objects and the spatial correct registration in 3D [2] with the potential for cognitive savings. Most context-aware assistance application methods are goal- [13] or attention-oriented [28] and therefore not presenting information about the way to solve the task, i.e. process-oriented [24]. Furthermore, mostly assistances are tailored to specific scenarios and not providing a general system for the development of context-aware assistances in different setups. Therefore, the adaption of those assistance systems and methods is not done in a large scale for the manual assembly, due to technical limitations [35]. While in the literature a simulation-based assistance, visualizing an augmented digital human during manual assembly tasks was proposed [24], it lacks on the one hand the ability to incorporate knowledge about context of the particular scenario through arbitrary sensor data besides spatial information, and on the other hand the extensibility and adaption of the system to other setups. Therefore, in this paper, besides the propose of a general framework for the development of context-aware assistances, the opportunities of the enrichment of context-awareness to a process-oriented human simulation based assembly assistance is subjectively investigated by a preliminary study.

2 Related Work

The developed assistance builds upon a large body of related work. In the following an overview concerning the related work of the encapsulation of simulation approaches and context-aware assistances during manual tasks is given.

2.1 Exchanging Simulation Approaches

The main idea behind exchangeable simulation approaches is to encapsulate specific behaviour of a complex system into several exchangeable standardized generic units. The general concept is derived from the Functional Mock-up Interface (FMI) [6] which is a standard utilized to simulate mechatronical components in a complex system such as vehicles. The proposed concept of encapsulated units was recently adapted to the domain of character animation. Hence, the Motion Model Interface (MMI) with the encapsulated Motion Model Units (MMUs) was presented [14]. By using encapsulated units together with a co-simulation process, various different character animation techniques and algorithms can be incorporated into a common system. For instance, for the simulation of a complex human movement separate MMUs for the modeling of different motion primitives, e.g. walk or grasp, can be combined so that a comprehensive human movement simulation is possible. The output can further be used as input for a human-simulation based assistance. Within this paper, the presented approach of encapsulated units is adapted for the interpretation of sensor data to context. Furthermore, the MMU approach has been utilized to generate the necessary movements of the displayed digital human.

2.2 Context-Aware Assistance

Since context-awareness is utilized to improve applications in different domains [13,16,19], the literature proposes various different definitions for the terms context-awareness and context. Due to the specific character of most definitions and consequently the hard applicability in multifaceted use-cases, the widely accepted context definition proposed by [1] is adopted. Within this definition there is no limitation of context with respect to specific parameters, but rather comprises any knowledge delineated the given status as context. Whereas, in terms of general context-awareness the authors [1] highlighted the importance of the user and the task to be performed, the dimensions of important context for assistances during manual tasks are extended to an environment model [7,10,20] and an interaction model [20,34]. Due to the variety of important context within the specific use-case of assistances during manual tasks, divers context-aware systems are developed and investigated. A wide range of assistances are developed in augmented reality (AR) and categorized as context-aware, due to the fact that AR applications are by definition context-aware, considering the spatial registration of AR content in the real world [15]. However, besides the spatial correct information presentation, there are more complex approaches, integrating algorithms and sensors for the utilization of additional context information and the associated adaption of the presented information. Within different work [5,23] system designs are presented, integrating motion recognition techniques to track body movements and thus, enabling implicit interaction with the systems by utilizing data from camera sensors. Thereby, context information with regard to the current task are concluded and an adaptive information presentation is ensured. Within experiments the developed systems are utilized and the potentials of different goal-oriented assistance methods are investigated [5,22]. Whereas the aforementioned systems are limited to context information derived by the capturing of body movement or object tracking, systems with additional user-centred functionality, e.g. by modeling the cognitive state of the user, are proposed [4,36]. Moreover, Westerfield et al. [36] proposed a modular system architecture to enable the possibility of an adaption to new task sequences, equally to [26,29]. Nevertheless, despite of the encapsulation of decision processes and the final visualization, the extensibility of the systems to integrate and utilize additionally context information is associated with great effort, due to the lack of a general context model. Furthermore, whereas mostly goal and attention-oriented assistance methods are enriched by context-awareness, a process-oriented human simulation-based assistance with the utilization of context information gathered by the Microsoft HoloLens for a step by step assistance was proposed [24]. However, other context-aware systems providing a process-oriented assistance with a digital human are mostly proposed as motor learning techniques in domains such as sport [9,19] and till now not adapted in assembly use-cases.

3 Context-Aware Assistance Framework

With regard to the presented related work, the main objective of this work is to develop a process-oriented context-aware assistance which enables implicit interaction with an augmented human. Therefore, a general encapsulated software architecture for the acquisition and management of context through the interpretation of sensor data is mandatory. Additionally, a base implementation of the framework is useful for the rapid modeling and development of context-aware assistance scenarios.

3.1 Overall Concept

To enable the general development of context-aware assistance systems, a novel concept for the standardized and encapsulated interpretation of sensor data to context, inspired by the presented related work, is proposed. In the following the context categorization of Hong et al. [21] is adopted. Hence, the data gathered by the sensors are further determined as preliminary context, whereas the term of interpreted context is defined as information deduced from preliminary context. Lastly, final context is the context integrated in a model and further sent to context-aware applications.

Context Interpretation Units. The integration of heterogeneous context sources is mandatory for a context-aware assistance. Only with various different sensor types the wide-ranging context of an assembly scenario can be gathered. To reuse developed context information independent of connected sensors within a common system, an encapsulation and standardization of this process is essential. To cope with the complexity of this task, the Context Interpretation Units (CIUs) were developed which have been highly inspired by the architecture and definition of the Motion Model Interface [14]. These units are generic black boxes, which use preliminary context from sensors and produce integrated context based on internal processing routines. By utilizing a suitable middleware technology the implementation of these standardized units are not restricted to one programming language or environment. For example, deep learning models created in Python as well as rule-based systems in C# can be utilized within the system simultaneously. Since these underlying processing mechanisms are not important for the overall framework, an encapsulation of recognition algorithms is consequently possible, if standardized input and output parameters as well as control signals are defined and known within the system (see Fig. 1).

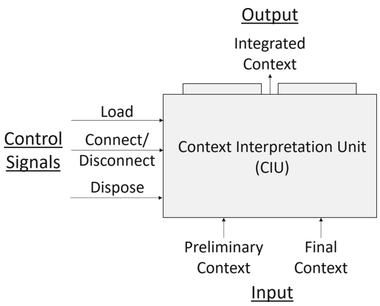


Fig. 1. Abstract illustration of the proposed Context Interpretation Unit (CIU).

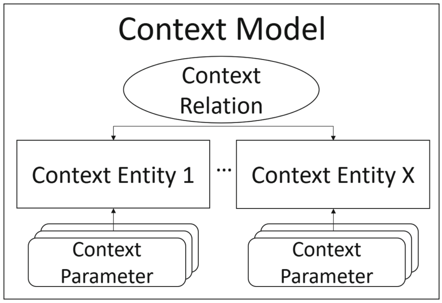


Fig. 2. Abstract illustration of the proposed context model.

Besides mandatory preliminary context by sensors, the input of final context can optionally be stated, since some sensor fusion or the actual recognition might be improved by context-awareness [32]. For example, the recognition of an object inside of a video might be improved, if the last position of it is known beforehand, due to a minimized search area. To control a CIU, independent of the actual implementation within the CIU, several control signals are mandatory. Before a CIU can start its recognition, it first has to load its dependencies and establish connections to the different source and target locations via the connect routine. If a CIU is not needed anymore, their connections can be closed using disconnect and any maintained resources are freed with dispose.

General Context Model. For the opportunity to infer higher-level knowledge, the derived information of the CIUs first has to be integrated into a given context model. To handle the various requirements of different context-aware assistance applications in manual task situations, the idea is to develop a general context model being capable of storing and providing divers context information to registered assistance applications. Therefore, the literature proposes a wide range of different context modeling approaches for a context information management [27]. However, the choice for an adequate approach depends on the specific use-case requirements. Considering the stated requirements of assistance systems for manual tasks an object-oriented approach was developed for the use-case of this work due to its encapsulation, reusability and inheritance functionality [33]. It further allows for complex modeling of relationships which is a huge advantage in the case of assistance systems. The presented approach consists of basic classes, here: context entities, characteristics, here: context parameters, and relations between individuals, here: context relations (see Fig. 2). A typical context entity could be the user itself, a sensor or a component needed for a given task. To further specify and describe a context entity, it can have an arbitrary amount of context parameters attached as attributes. According to the context atoms by [3], it incorporates a name, its value, the source, which detected its current value, and the time of the last detection. To allow the dealing with uncertainty a confidence value [0:1] is further added. In addition to context parameters, pairwise context entities can also be defined by context relations. With the possibility of the definition of context relations, real world relations between entities can be modeled (e.g. relationships between a carried object and the carrying user).

3.2 Technical Framework

The overall technical framework with its data flow, integrating the presented concepts of CIUs and the general context model, is shown in Fig. 3. As it can be seen the developed system consists of three different main parts, namely the context acquisition, the context management and the context-aware assistance application.

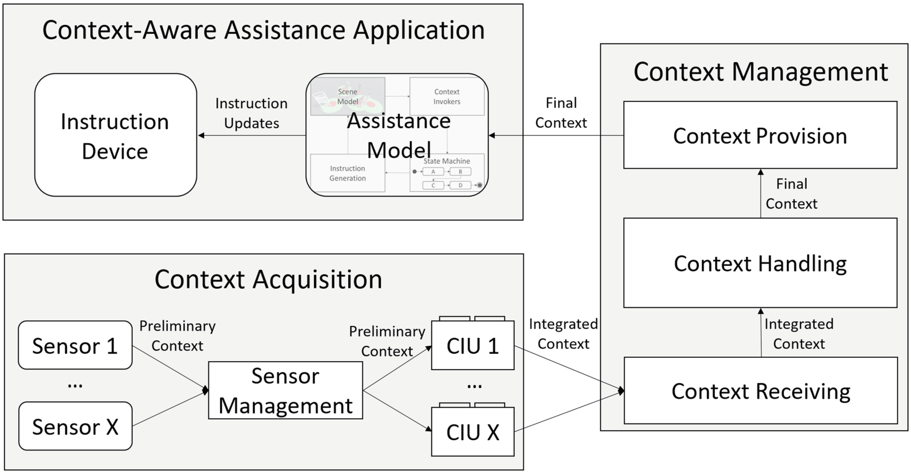


Fig. 3. Proposed software architecture of the context-aware assistance framework.

Context Acquisition. The integration and management of different sensor types, the transmission of preliminary and integrated context and the definition and function of CIUs are part of the overall context acquisition process.

First, the sensor management provides a Message Queue Telemetry Transport (MQTT) [17] broker so that the sensors can connect and deliver their measured preliminary context to it and subscribed clients. Such clients are the CIUs which can listen for distinctive data streams and process their data further; hence, various sensor types can be integrated. Second, the management of the connections is a crucial task of the sensor management. Whenever a sensor connects or disconnects due to failures, the sensor management is utilized to supervise these events and inform dependent processes.

To be able to control these CIUs some meta information about them is necessary for the framework, namely a general description of its behaviour and its input and output parameters. For the input and output parameters, the actual required sensor data and the defined integrated context parameters within the context model have to be known. For the description, a unique ID, the programming language of the unit as well as a dependency list and a confidence item have to be set. With every incoming input a new recognition is performed and the CIU decides whether an update is send. To have a uniform transmission format the updated values are provided by utilizing an integrated MQTT client and a JSON serialized version of the aforementioned context parameter. The incorporated sensors and CIUs in this work are implemented by using Unity 2018.3.2 and .NET Framework 4.6.1.

Context Management. The context management was designed in accordance with the proposed context server architecture by [8]. This enables the access of the modeled and interpreted context data for multiple applications and also moves often needed resource capacity for the context reasoning from the applications to the server. The context management architecture is further divided into three parts, namely the context receiving, context handling and context provision. The context receiving is on the one hand utilized to obtain integrated context of the CIUs by the provision of an MQTT broker endpoint, and on the other hand to manage the CIUs. For that, programming language dependent mediators inspired by the adapters in the MMU framework [14] have been developed which are able to start the CIUs of their language and connect them to their endpoints or disconnect and stop them respectively.

The context handling incorporates the integrated context gathered by the context receiving into a defined context model and processes it further through model related reasoning to higher-level final context. The interfaces for the items of the context model and reasoning were realized in C# (.NET Framework 4.6.1) and the actual base implementations of the classes were implemented within the Unity engine (Unity 2019.1.8) as scripting components. The reasoning interface contains a list of input context parameters and context relations. The base reasoning listens for context parameter changes but does nothing in case of an update. With this basic implementation, the development of sophisticated reasoning approaches based on rules, fuzzy-logic or machine learning should be possible. The usage of a game engine for the modeling and reasoning allows for the development of easy to use designing tools and the usage of GameObjects, their transforms and rendering leads to a better understanding of the overall scene relationships which is helpful especially for the designing of complex models. Relating thereto, since most manual task assistance applications addresses visual sensory modalities, the usage of a game engine has advantages.

After the evaluation of the higher-level context the resulting final context is send to registered assistance applications and CIUs as a serialized version of the updated parameters. The data is transferred by the integration of a MQTT broker of the context provision, likewise to the receiving.

Context-Aware Assistance Application. Similar to sensors, instruction devices have typically only a limited computation power so a client-server architecture via MQTT has been developed analogously to the sensor management. Therefore, the calculations of the process logic within the actual assistance is separated from the output device.

The assistance model receives the updated final context of the context management and includes the model logic of the context-aware assistance application. Respectively, it is responsible for the generation of the context-aware instructions. The assistance model is implemented with Unity 2019.1.8, and incorporates four core components, namely a state machine, scene model, context invokers and instruction generation. Whereas the states and transitions of the state machine are realized as scripting components which references can be set by the developer to certain instruction generation algorithms, the scene model is the actual Unity scene of the assistance. The scene model is updated by the output of the instruction generation algorithms. Hereafter, the resulting changes are send as instruction updates to the subscribed instruction device via MQTT as well as transferred as inputs to the context invokers. With the knowledge out of the updated scene model and the context management, predefined conditions within the context invokers can be evaluated and the state machine can be updated based on the obtained context-awareness. Depending on the assistance application, the instruction updates can be utilized to transfer any serializable object and information to subscribed listeners. Thus, with regard to the instruction updates no limitation considering the addressed stimulus modality of the instruction is given. The only requirements of the device are that it contains one or more MQTT clients which are connected to the broker of the assistance model and that routines are implemented to adapt their information output based on the information delivered by the instruction updates.

4 A Context-Aware Augmented Human Assistance

Besides the proposal of the general concept of an encapsulated architecture for context-aware assistances, the aim of this work is to apply the aforementioned architecture and develop an assistance which enables implicit interaction with an augmented human during the completion of manual tasks. Therefore, based on the underlying related work, we want to enhance the assistance approach presented by [24] due to the incorporation of knowledge derived by various sensors as well as the consideration of social interaction guidelines for the development of additional context-aware features. Therefore, the gap between context-awareness and process-oriented assistances for manual assembly use-cases is addressed.

4.1 Overall Implementation

The stated technical framework with its underlying concepts is utilized for the implementation of the context-aware augmented human assistance. The three main parts of the proposed framework are adapted to the specific requirements.

Context Acquisition. Considering the stated architecture of the context acquisition, two different types of sensors are exemplary connected within the context-aware augmented human assistance, namely the Microsoft Kinect 2 and the Microsoft HoloLens. Therefore, on the one hand, area observations enabled by the depth sensor data of the different Kinects [23] and on the other hand, positional and rotational tracking of the user by the inertial measurement unit data of the HoloLens are available. More precise, the related C# and Unity CIUs supply integrated context of the global position and rotation of the user within the Unity scene of the context model and furthermore, integrated context of changes in specified point of interests (POIs) within the context-model.

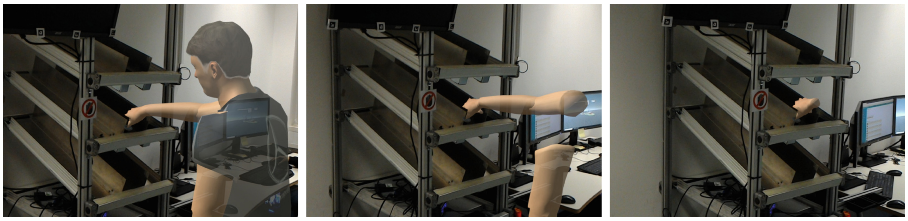
Context Management. To integrate the provided context information, the context management comprises the specific context model with its context entities, parameters and relations. Furthermore, adapters have been developed for the currently utilized Unity and C# CIUs. Within the Unity scene a true to scale reflection of the laboratory setup (see Fig. 6) is available. To go into detail, besides the CAD models, a sensor context entity was modeled for each connected sensor. Furthermore, for each POI a context parameter was modeled and added to its corresponding Kinect sensor entity. Similarly, for the position and rotation of the user a context parameter was added to the HoloLens sensor entity, as well as a context parameter of the related view frustum with the frustum quantities of the HoloLens. Moreover, for the matching of the context model to the real world an AR mapping entity was modeled. With the implementation of certain context reasoners higher level context can be utilized. For example, if the parameter of a POI of the rack changes and the user is near the POI a context relation between the user and the part related to the POI is updated.

Context-Aware Assistance Application. With regard to the context-aware assistance application of the technical framework an instruction device as well as the four belonging components of the assistance model are utilized and adapted to the needs of an augmented human assistance application. With regard to the suggestion of a hands-free usage during manual tasks [13], the head-mounted display (HMD) HoloLens is utilized as instruction device. Furthermore, to display an augmented human on the HoloLens an Universal Windows Platform application was created. The scene contains the same digital human and digital component objects as the assistance model. Using initially Vuforia 8.3.8 the position and rotation of an image target, are placed correctly in the real world by updating the AR mapping entity of the context model and therefore, the digital components overlap the real ones. First of all, to be able to assist a user, an assistance sequence has to be modeled utilizing the base implementation of the aforementioned state machine component. For the generation of the instruction, i.e. the movement of the augmented human, the modular MMU framework proposed by [14] is incorporated. Within each state of the state machine the simulation process of the subsequent human simulation data is started or stopped by the control signals passed to the MMU framework. Within the incorporation of the proposed MMU framework, the feasibility of MMUs with pre-captured or pre-simulated motion data, as well as model-driven and data-driven simulation approaches is given. As an output of the states the human simulation data, comprising the updated pose of the avatar and the manipulated digital components, are further incorporated into the scene model and send to the subscribed HoloLens. By utilizing the functionality of context invokers, context-aware features can be realized.

4.2 Context-Aware Features

With regard to social interaction guidelines in AR and by utilizing the stated implementations, implicit invoked context-aware features enhancing the assistance approach of [24] are realized. More precise, four features are integrated in the overall context-aware augmented human assistance, namely the visibility control, progress control, attention control and feedback control.

Visibility Control. A controllable parameter that can enhance the effect of the assistance, is the visibility of the task performed by the augmented human. On the one hand, it is important that the body of the avatar is not obscuring important information and on the other hand, it is important to reduce the discomfort of being inside the avatar's proximity as mentioned by [25]. Due to the updated context information of the user's position, derived by the HoloLens and the updated avatar's position, a predefined condition inside the context invoker can be evaluated at each frame. With consideration of the work on proxemics [18], the implicit visibility control is enabled (see Fig. 4). If the avatar is in the personal space of the user ($\leq$1.20 m), only the avatar's arms are visualized, whereas within the intimate space ($\leq$0.46 m), only the hands are displayed.

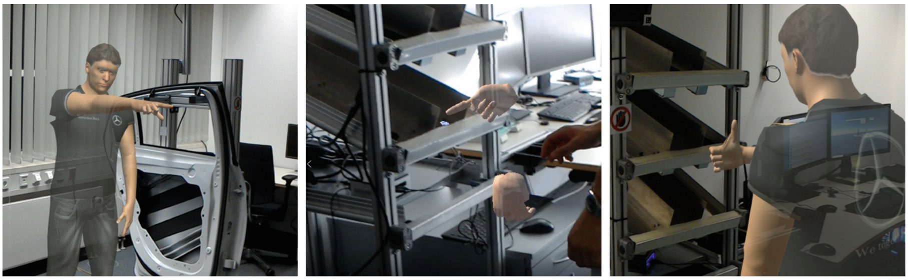


(a) (b) (c)

Fig. 4. Exemplary visualization of the proposed implicit visibility control feature that displays (a) the full body, (b) arms and hands and (c) only the hands of the augmented human.

Progress Control. The conformance between the speed of the assistance and the individual character of the user performing a manual task, considering mainly the speed, has an important impact on the assistance's outcome [13]. If the user is spatially or temporally far behind the augmented human, the possibility exists that the user misses important actions. Therefore, the context-aware feature of progress control is implemented, whereby the augmented human waits if the eventuality of information loss exists. A basic chapter functionality is enabled by the evaluation of the distance between the avatar's end position after locomotion tasks and the user's position. For manual tasks with an environment interaction, the provided final context of the observations of the POIs triggers the subsequent movement of the augmented human. Therefore, a more sophisticated chapter functionality besides the basic implementation is integrated.

Attention Control. An attention control feature is added, with regard to the limited field of view (FOV) of the AR device. To minimize searching movements, occurring because the FOV of the HMD might not be superimposed with the POI, pointing movements of the augmented human are implemented. For one thing, the attention guidance is triggered during walking tasks, when the user searches the augmented human and thus looks at a wrong direction, inferred by the final context of the HoloLens sensor entity (see Fig. 5(a)). During manual tasks with an environment interaction, attention guidance is enabled, if a manipulation of the environment at a wrong POI is detected by the Kinects (see Fig. 5(b)). Therefore, if a lack of orientation is recognized based on the aforementioned conditions, the attention control feature is invoked.



(a) (b) (c)

Fig. 5. Exemplary visualization of the proposed implicit (a) attention control during walking tasks, (b) attention control during manual tasks and (c) feedback control features.

Feedback Control. With respect to the guidelines of [13], the motivational state of the user is important regarding the assistance outcome. Thereby, the user's knowledge of the correct completion of the manual tasks with its subdivided chapters is important. As a result of the environment manipulation information derived by Kinects and the knowledge of the predefined task sequence the task completion condition can be evaluated. Via a thumbs-up gesture of the augmented human (see Fig. 5(c)) the user is aware of the completion of particular chapters.

5 Preliminary Study

Regarding the evaluation of the proposed context-aware augmented human assistance with the adaption of the general assistance framework, a preliminary study was conducted. Within the study we investigated the user experience and the subjective perception of the developed context-aware features in terms of time, error and cognitive load reduction, as well as motivational increase.

5.1 Experimental Design

A within-subject study design was utilized with one independent variable with two factor levels, i.e. the different assistance methods. As the baseline method the simple simulation-based assistance by [24] was used and compared against the presented context-aware adaptive simulation-based assistance. With respect to the comparability of the two methods and the definition of context-awareness in AR [15], the basic chapter functionality of the progress control evoked by the distance between the user and the avatar is utilized in all methods as a trigger for the subsequent task. Therefore, the simple simulation and the context-aware adaptive simulation differ respectively by one variable, i.e. context-awareness; whereby insights of the effects can be revealed. Three different manual task sequences, following real-world scenarios, were constructed during the experiment. The sequence of the tasks, each consisting of picking, carrying, plugging and screwing, were the same in all assistances to ensure the same complexity for each method. Furthermore, we counterbalanced the order of the factors across the participants. Besides the measurement of the user experience, obtained by the UEQ-S [31], the participants were asked to rate the four context-aware features in terms of error savings, time savings, cognitive savings and increased motivation.

5.2 Apparatus

For the experiment a laboratory door assembly setup was utilized, covering an environment of $6\,\mathrm{m} \times 7\,\mathrm{m}$ (see Fig. 6). In the setup, a start zone, picking zone, pre-assembly zone and assembly zone exist. Every zone, except the start zone is equipped with a Kinect. Both sides of the (pre-)assembly zones were used. The rack within the picking zone comprises eleven compartments. With regard to the AR information provision, we utilized a HoloLens as the instruction device for the assistance methods. On the basis of an increase of acceptance due to natural motions, only motion data captured with the XSens Motion Capture system [30], stored in MMUs are presented.

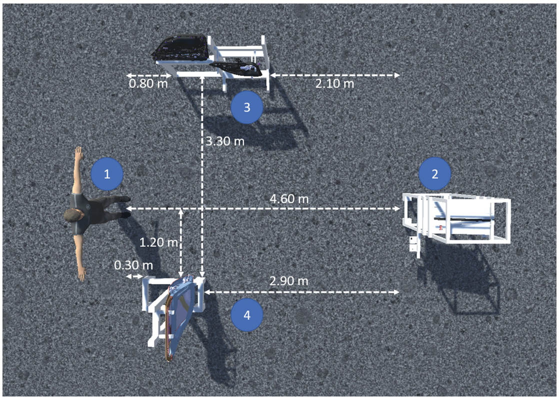


Fig. 6. The visualized constructed work station in Unity containing the start/end zone (1), picking zone (2), pre-assembly zone (3) and assembly zone (4).

5.3 Procedure

First of all, the participants were welcomed by the experimenter. Afterwards they had to sign a declaration of consent as well as complete a demographic questionnaire. Hereinafter, the course of the study was described. It was explained that the participants had to assemble three tasks with their subtasks while they are guided by a worker assistance in AR with the HMD. Moreover, the participants got used to the different questionnaires. To make the participants familiar with the assistance method and the hardware, they had to conduct a short test task. This test task was the same for both assistance methods. During this task all features of the different assistance methods were named and it was ensured, that all features were seen by the participants. After the fulfilment of the test task the participants were asked, whether they would like to repeat this task or start with the actual experiment. It was further clarified that no further questions can be answered during the main experiment. For the start of each of the three tasks the participants were asked to stay at the starting position and to look at the rack. The experimenter started the assistance method remotely as soon as the participants communicated their readiness. Afterwards, the UEQ-S had to be completed and the procedure was repeated with the next assistance method. At the end, the context-aware feature questionnaire was answered by the participants. Overall, the study took approximately 50 min.

5.4 Participants

Overall 6 participants, students and research engineers, took part in the within-subject experiment. The participants (2 female, 4 male) were aged between 24 and 29 (M = 26.16 years and SD = 1.77 years). Prior knowledge of the assembly tasks were not present for any participant. Furthermore, they did not get any extra rewards for their participation.

5.5 Results

A statistically comparison by considering M and SD of the two assistance methods was conducted for the resulting UEQ-S scores and the Likert scale ratings. Furthermore, for interference statistics the Shapiro-Wilk test as well as the Levene's test proved existences of normal distribution and variance homogeneity. Therefore, a paired sample t-test was utilized to test for significance. Finally the effect size with Pearson's r was quantified with 0.10 for a small, 0.30 for medium and 0.50 for large effects [12].

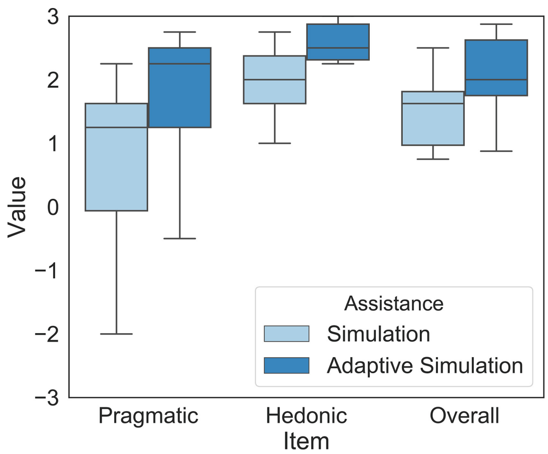


Fig. 7. Box-plot with the resulting pragmatic, hedonic and overall UEQ-S scores for the two assistance methods.

User Experience. For the user experience, the pragmatic and hedonic quality values together with the overall ratings of the methods are shown in Fig. 7. As can be seen, the adaptive simulation was rated to have the greatest pragmatic quality with $M = 1.71$ and $SD = 1.14$, greatest hedonic quality with $M = 2.38$ and $SD = 0.67$ and therefore the greatest overall quality with $M = 2.04$ and $SD = 0.68$. It is followed by the simulation approach which pragmatic quality was rated with $M = 0.67$ and $SD = 1.46$, hedonic quality with $M = 1.96$ and $SD = 0.58$ and overall quality with $M = 1.31$ and $SD = 0.96$. Furthermore, for the overall quality interference tests were conducted. The paired t-test proved that the results are not significant with $p \geq 0.05$ for all three quality values. However, a large effect size $(r_{pragmatic} = 0.98, r_{hedonic} = 0.57, r_{overall} = 0.90)$ can be observed.

Context-Aware Features Ranking. In Fig. 8 the results for the different context-aware features are visualized. As it can be seen, the progress control was perceived as the most useful in terms of error reduction with $M = 4.83$ and

SD = 0.37 (visibility control: 4.33 ± 1.11, attention control: 3.17 ± 1.86, feedback control: 3.17 ± 1.57). In comparison, in terms of time reduction the attention control with M = 4.33 and SD = 0.75 is rated as the most useful function (visibility control: 3.33 ± 1.11, progress control: 3.17 ± 1.07, feedback control: 2.33 ± 1.60). Furthermore, the cognitive load reduction was rated similar to the error reduction with the progress control functionality as the most useful with M = 4.50 and SD = 0.76 (attention control: 4.00 ± 1.41, visibility control: 3.50 ± 1.50, feedback control: 3.33 ± 1.49). On the other hand, the feedback gesture was rated as the most useful means to increase the motivation with M = 4.17 and SD = 0.90 (progress control: 3.67 ± 0.94, attention control: 3.17 ± 1.34, visibility control: 2.33 ± 1.37).

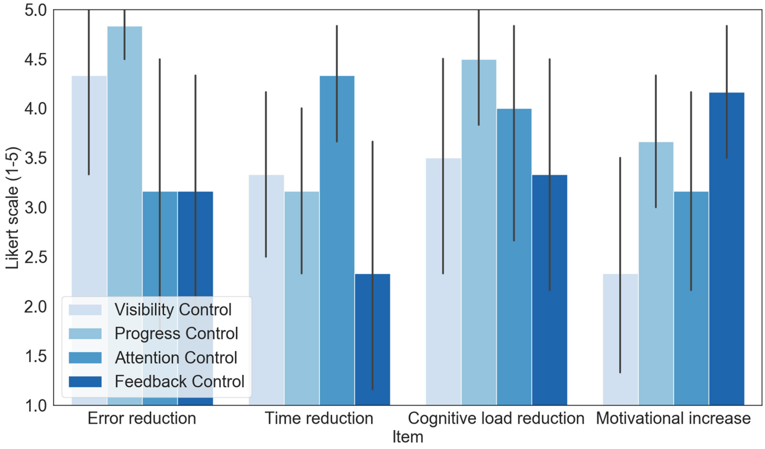


Fig. 8. Barplot with the resulting rankings (5 point Likert scale) of the implemented context-aware features with respect to the four evaluation criteria.

5.6 Discussion and Limitations

Considering the evaluation criteria, the descriptive results demonstrate that the context-aware adaptive simulation improved the user experience in comparison to the baseline method. Whereas both simulation approaches are perceived to have considerable high pragmatic, hedonic and overall quality, the results further suggest that the enrichment with contextual knowledge is an influencing factor with a large effect.

With respect to the underlying industrial use-case of assistance for manual assembly, especially the importance of the pragmatic quality should be taken into consideration. The positive pragmatic quality results could be explained by the perceived cognitive load. The less challenging a task is, due to cognitive load reduction by the assistance, the more practical this system might be with regard to performance parameters such as made errors and needed time. Considering the subjective rating, especially the progress and attention control lead to cognitive savings as well as error and time reduction.

The positive hedonic values for both assistances can be further explained by the participants familiarity with the respective assistance method. It can be assumed that the participants are familiar with situations of motor learning by observation in real-world scenarios, however not in augmented reality. Furthermore, current AR assistance methods mostly utilize small holograms, due to the restrictive FOV of AR devices, whereas the augmented human is normal sized with a body height of 1.75 m. Hence, the ranking of the attention control feature especially reveal positive effects with regard to the needed time and cognitive load savings due to a decrease of searching movements. Therefore, the results suggest, that the context-awareness is especially beneficial with regard to the small FOV of current AR HMDs. Nevertheless, further investigations of the influence of the FOV associated with the context-aware features are crucial.

With regard to the implemented context-aware features, especially the visibility control feature appeared as very helpful during the experiment, due to the fact that for the simple simulation method some participants walked around the avatar instead of stepping inside and therefore missed some information. This effect of the visibility feature can be illustrated by taking the ranking for the error reduction into account. Future research with an objective measured quantity is mandatory to verify the assumption.

Overall the subjective ranking of the features demonstrate in particular the effectiveness with regard to error and cognitive load reduction, whereas the time reduction is only rated above 3.5 for the attention feature. Thereupon, it can be inferred that the presented context-aware assistance with its implemented features take effect especially in learning situations or complex situations, where the need of free attention capacity exists and time saving is less important than faultlessness. However, with the possibility of extensibility and adaptability of the proposed system architecture, other features with the aim of time saving could lead to other results. Summarized, the integration of specific context-aware features depends on the particular requirements of the use-case.

Generally, with respect to the results, the preliminary study indicates that within a laboratory setting an process-oriented simulation-based assistance approach can be enhanced by the enrichment of contextual knowledge with regard to subjective perception. However the results should be verified within a user study with a respective number of participants and the consideration of performance criteria, so that significant results can be revealed. But with the study it was shown, that the overall architecture can be used for context-aware assistance systems in a stable and effective way. During the entire experiment a steady information presentation was given. Furthermore, the proposed overall system architecture was used for both assistance methods, utilizing a different quantity of context information. Altogether, the adjustability of the overall system is given. However, the developed software architecture should be evaluated with the integration of more sensors and more complex context interpretation algorithms. Moreover, the framework offers potential for improvement considering the strong dependencies between the assistance model and the instruction device.

6 Conclusion and Future Work

Based on previous work, the main objectives of this paper was to enhance a simulation-based assistance by incorporating contextual knowledge from sensors. For that, a concept and subsequent implementation of a general context-aware assistance framework was presented. This system facilitates the modular interpretation of sensor data and encapsulates the actual assistance application from the context acquisition and management with the result of a faster development of context-aware assistances, incorporating contextual and virtual scene information. Utilizing the proposed architecture, the previously proposed process-oriented simulation-based assistance was enhanced. The approach is able to adapt its simulation scheme with several context aware features, enabling an implicit control of the visualized augmented human. Therefore, the proposed system in general addresses the open gap between process-oriented assistances and the enrichment of context-awareness.

Within a preliminary study the effect of the developed assistance method with its context-aware features was evaluated subjectively in comparison to the baseline simulation. While no significant results could be measured, probably due to the small amount of participants, the effect size is revealed as large and the results therefore suggest a general validity. Considering the results, it can be seen that the user experience of the adaptive simulation method is rated higher than the baseline simulation. The context-aware features of the adaptive simulation method have an impact especially on the pragmatic value, which is an important factor for assistances and also might have an impact on the perceived cognitive load. Furthermore, the context-aware features were overall perceived as useful in terms of error, time and cognitive load reduction as well as motivational increase. Nonetheless, the proposed approach enhances the baseline simulation-based assistance only subjectively. A user study with more participants and the consideration of performance criteria is mandatory so that the measurable effects of the adaptive simulation can be quantified. Additionally, the proposed concepts, software architecture and implemented assistance have still the potential for improvement. Firstly, the context interpretation units are currently limited to the utilized context information and sensors of this work. By incorporating different sensors and implementing algorithms to interpret their data, other context information could be acquired and consequently additional context-aware features could be created and evaluated for an effective adaptive simulation-based assistance. Secondly, in this work only one context-aware assistance application was running at the same time, due to a close connection as one-to-one relationship between acquired context and utilized assistance application. This connection might be encapsulated by a modular structure for assistance applications through appropriate models and exchangeable data formats. Thus, several other assistances could be easily integrated into the system and started simultaneously, so that a more holistic guiding of a worker is possible.

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